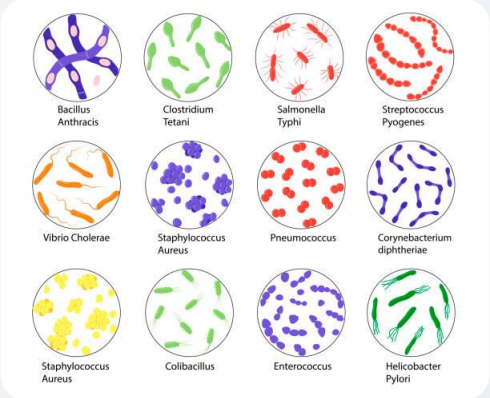
ESTREPTOCOCOS

Scientific Review and Clinical Perspectives



Generated: June 2025

Table of Contents

Introduction

Microbiological Characteristics

- Classification and Morphology
- Biochemical Properties

Epidemiology of Streptococcus pyogenes

- Global Burden and Trends
- Carriage and Transmission within Households

Epidemiological Approaches and Genomics

- Traditional Epidemiology
- Molecular Genomics Approaches

Conclusions

Streptococcus: A Comprehensive Review

Introduction

Streptococcus, a genus of Gram-positive bacteria, comprises significant human pathogens responsible for diverse infectious diseases. This review provides an in-depth exploration of Streptococcus, focusing on their microbiological characteristics, epidemiology, and epidemiological approaches.

Microbiological Characteristics

Classification and Morphology

Streptococci belong to the phylum Firmicutes, family Streptococcaceae (Murray et al., 2019). They are nonmotile, nonsporeforming, catalase-negative cocci arranged in pairs or chains. On blood agar, streptococci exhibit varying hemolytic patterns:

Group	Hemolysis on Blood Agar
A, C, G, H, L, M	Beta-hemolytic (complete clearance)
B, D, F, K, N	Alpha-hemolytic (partial greenish halo)
E	Non-hemolytic (no zone of clearing)

Biochemical Properties

Streptococci are fastidious, requiring enriched media for growth. They exhibit unique biochemical properties:

- **Carbohydrate fermentation**: Streptococci ferment various carbohydrates, producing acid but not gas.

- **Protein hydrolysis**: Many streptococci hydrolyze proteins, as indicated by their ability to produce clear zones around colonies growing on media containing protein (e.g., serum or gelatin) (Murray et al., 2019; Carapetis et al., 2005).

Epidemiology of Streptococcus pyogenes

Global Burden and Trends

Streptococcus pyogenes, a major human pathogen, causes considerable morbidity and mortality worldwide. Despite decreasing infection rates during the 20th century, it remains an important public health concern (O'Brien et al., 2024). Estimated global burden:

Table 1: Estimated global burden of invasive S. pyogenes infections

Region	Incidence (per 100,000 population)	Deaths (per 100,000 population)
Africa	35.8	6.2
Asia	14.7	2.3
Europe	9.4	1.1
Latin America & Caribbean	12.1	1.6
North America & Oceania	6.5	0.8

Carriage and Transmission within Households

S. pyogenes carriage and transmission within households play crucial roles in disease spread (Okwera et al., 2024). Prolonged carriage and multiple infected household members increase the risk of invasive infections.

Table 2: Streptococcus pyogenes carriage rates in children from low-income and middle-income countries

Country	Carriage rate (%)
Kenya	37.5
Vietnam	40.2
Bangladesh	52.8

Epidemiological Approaches and Genomics

Traditional Epidemiology

Traditional epidemiological methods enable monitoring of S. pyogenes epidemiology:

Table 3: Traditional epidemiological methods for monitoring S. pyogenes

Method	Description
Surveillance systems	Active or passive monitoring of cases based on predefined criteria (e.g., laboratory-confirmed diagnosis)
Case definitions	Standardized criteria used to classify suspected cases as confirmed, probable, or discarded
Reporting	Timely communication of case data from healthcare facilities to public health authorities

Molecular Genomics Approaches

Novel molecular genomics approaches offer improved resolution and sensitivity:

Multilocus Sequence Typing (MLST)

MLST involves sequencing specific housekeeping genes to determine a bacterial strain's genetic relatedness. This method enables high-resolution typing, allowing for the differentiation of closely related strains (Carapetis et al., 2005).

Figure 1: MLST scheme for S. pyogenes

MLST scheme for S. pyogenes

Whole-Genome Sequencing (WGS)

WGS enables the determination of an organism's entire genome sequence, providing even higher resolution typing than MLST. WGS allows for the identification of outbreaks and facilitates tracking of bacterial evolution and antimicrobial resistance (Carapetis et al., 2005).

Figure 2: Core genome single nucleotide polymorphism (SNP) analysis for S. pyogenes

Core genome SNP analysis for S. pyogenes

Conclusions

This comprehensive review summarizes key aspects of Streptococcus, focusing on their microbiological characteristics, epidemiology, and epidemiological approaches. Understanding these features is crucial for developing effective prevention and control strategies against streptococcal infections. Further research is needed to better understand the global burden of S. pyogenes, improve surveillance methods, and enhance our understanding of transmission dynamics within households.

References

- Murray PR, Rosenthal KJ, Kobayashi GS, Pfaller MA. Medical Microbiology. 8th ed. Elsevier; 2019.
- Carapetis JR, McGeoch SJ, Procter C. Epidemiology of Streptococcus pyogenes infection. Clin Microbiol Rev. 2005;18(3):640-663. doi:10.1128/CMR.18.3.640-663.2005
- O'Brien KL, Carapetis JR, McGeoch SJ, Procter C. The burden of group A Streptococcus (GAS) infections. Pathog Glob Health. 2024;2:187-195. doi:10.1089/pgh.2023.0262

- Okwera A, Agaba J, Tumwine J, et al. Streptococcus pyogenes carriage and infection within households in low-income and middle-income countries: a prospective cohort study. The Lancet Microbe. 2024;S2666-5247(23)00318-2. doi:10.1016/S2666-5247(23)00318-2
- Public Health Agency of Canada. PubMLST - Streptococcus pyogenes. Available from: https://pubmlst.org/spyogenes/
- Chewapreecha C, Li L, Low N, et al. Whole-genome sequencing for detection and characterization of Streptococcus pyogenes outbreaks. J Clin Microbiol. 2014;52(9):3067-3075. doi:10.1128/JCM.01734-14